PALETTE: Partition-based Affine Local-intEnsity Transform for Texture-prEserving augmentation

Anonymous CVPR submission

Paper ID *****

Abstract

Segmentation models for medical images are trained on one acquisition setting and deployed on others, and label-generative domain randomization (SynthSeg) addresses this by resampling intensities from a per-label Gaussian mixture over a dense anatomical label map – an approach that is both label-driven (it requires exactly the expensive annotation the method is trying to reduce reliance on) and texture-destroying (it replaces real gradients and partial-volume detail with piecewise noise). We introduce PALETTE (Partition-based Affine Local-intEnsity Transform for Texture-prEserving augmentation), a single-source, label-free augmentation that partitions the foreground of a scan into intensity classes with unsupervised k-means, further subdivides each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap $y = \mu + \alpha(x - \bar{x})$, $\alpha \in \pm U(0.5, 2.0)$, to the real *intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail are preserved within each sub-region; only the region-to-region contrast relationship is randomized. Training a single nnU-Net configuration with PALETTE and evaluating across eight cross-contrast and cross-modality settings spanning abdominal CT/MRI, brain MRI, and brain tumor MRI, we show significant gains over a no-augmentation baseline and the original label-generative SynthSeg in all eight settings, and over a stronger SynthSeg variant we re-implement in seven of eight. Against Auglab, a recent published augmentation pipeline that is the strongest competitor and that our full method builds on, adding the PALETTE transform improves all eight settings and does so significantly in six. A causal ablation that holds the partition and per-region statistics fixed while replacing the real-intensity remap with label-style noise fill isolates texture preservation, rather than the shared partitioning scheme, as the source of the advantage.*

1. Introduction

Medical image segmentation models are trained on one acquisition setting and routinely deployed on another: a different scanner, sequence, or imaging modality than the one seen during training. Because MRI contrast is a protocol choice rather than a physical constant, and because CT, ultrasound, and MRI encode the same anatomy in incompatible intensity distributions, a segmentation network that has only ever seen T1-weighted brain images can fail outright on T2-weighted or FLAIR images of the same anatomy, let alone a different modality. Collecting and annotating training data for every contrast and modality a model might encounter in deployment is not practical: manual segmentation is expensive, and clinical sites rarely share raw data across institutions. The single-source domain generalization (DG) setting — train on one labelled dataset in one contrast, generalize to unseen contrasts and modalities without any target-domain data — is therefore the setting that matters for real deployment, and it is the setting we address.

The dominant single-source strategy is label-generative domain randomization, popularized by SynthSeg [3]: given a dense anatomical label map, intensities are resampled from a per-label Gaussian mixture, producing images of arbitrary, unrealistic contrast that force the network to ignore contrast and rely on shape and layout. This is effective, but it comes with two structural costs. First, it is *label-driven*: it requires a dense anatomical segmentation of every training subject to resample from, so generating synthetic contrasts costs exactly the expensive annotation the method is trying to reduce reliance on. Second, and less discussed, it is *texture-destroying*: replacing every labelled region with piecewise Gaussian noise erases the real gradients, partial-volume boundaries, and unlabelled fine structure that the source image contains. Layout is preserved; texture is not.

We ask whether texture-destroying synthesis is necessary for contrast robustness, or merely the path of least resistance once a label map is assumed. We introduce **PALETTE** (*Partition-based Affine Local-intEnsity Transform for Texture-prEserving augmentation*), a single-source augmentation that requires no label map at all. PALETTE partitions the foreground of a scan into intensity classes with unsupervised 1-D k-means, further subdivides each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap $y = \mu + \alpha(x - \bar{x})$, $\alpha \in \pm U(0.5, 2.0)$, to the *real* intensities within each sub-region. Because the transform remaps existing intensities rather than replacing them with sampled noise, edges, gradients, and partial-volume detail are preserved within each sub-region; only the region-to-region contrast relationship is randomized, including sign inversions that no physical acquisition protocol produces. The result is a label-free augmentation that manufactures large, systematic contrast diversity while leaving the texture cues a segmentation network can exploit intact.

This distinction — image-driven, texture-preserving remapping versus label-driven, texture-destroying resampling — is not a matter of degree: the two families make opposite bets about what a network should be invariant to. We test this bet directly. Training a single nnU-Net [12] configuration with PALETTE, tuned once and reused unchanged across every dataset, we evaluate on eight cross-contrast and cross-modality settings spanning abdominal CT/MRI (CHAOS → AMOS/SLIVER07), brain MRI (ON-Harmony, Open-MS), and brain tumor MRI (Brats2024-GLI), against a no-augmentation baseline, both variants of SynthSeg (the original, label-generative noEM variant, and an EM variant for which no public implementation exists and which we re-implement), and Auglab [20], a recent published augmentation pipeline that is the strongest competitor. PALETTE outperforms the no-augmentation baseline and SynthSeg-noEM by a large margin in all eight settings, and beats the stronger SynthSeg-EM baseline significantly in seven of eight. Because our full method applies PALETTE on top of the Auglab pipeline, the comparison against Auglab isolates exactly the PALETTE transform: it improves aggregate Dice in all eight settings and significantly in six. The margin on aggregate whole-organ Dice is often thin — the setting is genuinely hard and Auglab is already strong — but it is large precisely where the texture argument predicts: on fine, texture-defined structures and on the most contrast-shifted test modalities, where destroying real texture costs the most (e.g. MS lesions in FLAIR, +8.9 Dice).

A thin average is not evidence for a mechanism; a mechanism claim needs a controlled test. We isolate texture preservation as the causal factor, rather than an incidental correlate of PALETTE's partitioning scheme, with a causal ablation: we hold the k-means/Voronoi partition and the per-region (μ, α) statistics fixed, and replace only the real-intensity affine remap with label-style noise fill, matching SynthSeg's resampling within PALETTE's own geometry. Structures that are texture-defined collapse under this ablation while boundary-defined structures do not, isolating texture preservation — not the partitioning scheme it happens to share with SynthSeg — as the causal source of PALETTE's advantage.

In summary, we make the following contributions:

- **PALETTE**, a label-free, single-source contrast augmentation that partitions a scan by intensity and space and applies a signed affine remap of the real intensities per sub-region, preserving texture and gradients that label-generative synthesis destroys.
- A systematic evaluation across eight cross-contrast and cross-modality segmentation settings (abdominal CT/MRI, brain MRI, brain tumor MRI) with a single, untuned-per-dataset configuration, against a no-augmentation baseline, both variants of SynthSeg (including a re-implementation of the SynthSeg-EM variant for which no public code exists), and Auglab, a recent published augmentation pipeline.
- A causal ablation isolating texture preservation, rather than the shared intensity/spatial partitioning scheme, as the source of PALETTE's advantage on texture-defined structures.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation (UDA), which assumes unlabelled target images and solves an easier problem [25]. Most medical DG methods are *multi-source*, interpolating across several labelled domains by meta-learning [17], feature alignment [29], or frequency-space exchange [19]. Recent computer-vision entrants follow the same recipe: MADGNet couples a multi-frequency architecture with multi-domain training [21], and foundation models such as SynthFM pursue modality-agnostic segmentation at scale [24], while diffusion-based stylization normalizes appearance but relies on a generative prior and on source diversity [2]. We instead work in the *single-source* regime — one labelled dataset, no target data, no second source — which cannot exploit cross-domain interpolation and makes architecture-coupled methods orthogonal: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [12].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family carries two structural costs. It is *label-driven* — a dense anatomical label map is required per subject, and SIAM shows the dependence is on label *quality*, needing a few high-grade manually curated templates [28]; producing such maps is expensive in expert hours. It is also *texture-destroying*: per-label Gaussian sampling replaces real gradients, partial-volume detail and unlabelled structure with piecewise noise, retaining layout but not texture. The lineage has been specialized to white-matter lesions [15] and MS lesions [4], and sits alongside training-free per-scan generative fitting such as SAMSEG [5]. We take SynthSeg's original (noEM) variant — which requires a dense anatomical label map — and its EM variant — which does not need a dense map but still uses the sparse task labels naturally available in a segmentation dataset to estimate its per-class intensity model — as baselines; as the EM variant has no public code, a faithful re-implementation is itself a contribution.

2.3. Single-source intensity and appearance augmentation

A second line manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [12, 20] is one such GPU-accelerated framework and is the strongest competitor in our experiments, and BigAug follows the same stacked-transform philosophy [33]. Our method inserts the PALETTE transform as one additional operator inside this pipeline, so the comparison against Auglab directly isolates the contribution of PALETTE. Closer to our operator are random-filter methods: GIN applies randomly re-initialized convolutions per forward pass to produce label-free appearance variants in true 3D with no trained parameters [22], RandConv establishes the principle for natural images [32], SLAug perturbs intensities per class using the task mask [26], and diffusion variants add learned generators [6]; a parallel frequency-domain line perturbs the Fourier amplitude spectrum rather than the image itself [18, 23, 34]. These share our label-free, single-source, architecture-agnostic stance — GIN especially is a direct peer — but vary appearance globally, spatially or spectrally, whereas our operator partitions the foreground into intensity classes and Voronoi subregions and applies a signed affine remap of the *real* intensities within each. The closest peer is semantic-aware random convolution [27], which also localizes the perturbation region-wise in true 3D, but differs on both axes that define our operator: it delineates regions from the *ground-truth label map*, reintroducing the task-label dependence we avoid, and applies a random convolution within each, which alters local gradients, whereas we partition label-free by k-means and Voronoi partition and apply an affine remap that leaves gradients intact. Because the remap is affine and region-local, edges and texture survive while contrast is randomized, preserving exactly the signal label-generative synthesis discards, without any anatomical labels.

2.4. The evaluation gap

We are not aware of prior work reporting our protocol: training on a *single real input modality* and testing on held-out modalities of the *same task and dataset*, across a benchmark spanning brain, abdominal and tumour anatomy in both MRI and CT. Existing single-source evaluations are either cross-dataset but same-modality — e.g. FLAIR MS-lesion models across sites [1] on cohorts such as the public 3D-MR-MS data [16] — or contrast-agnostic-by-synthesis, where training data already span all contrasts and "single-modality" refers only to the test input. The one-real-modality-in, held-out-modality-out axis is the harder setting, and the one we target.

3. Method

We describe PALETTE (§3.1), how it is applied during training (§3.2), and the causal ablation used to isolate texture preservation as the source of its effect (§3.3).

3.1. The PALETTE transform

Let x denote the intensity volume of a training scan, min-max normalised to $[0, 1]$, and $F \subset \Omega$ its foreground voxels (an intensity threshold; $x_v > 0.01$ in our implementation). PALETTE applies four steps at every training iteration, illustrated end-to-end in Fig. 1 on one real CHAOS (abdominal MRI) and one real OnHarmony (brain MRI) volume.

1. Intensity partition. We run 1-D k-means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$ (Fig. 1, column 2). This partition is unsupervised and requires no anatomical label map: it groups voxels by intensity alone, so the same operator applies unmodified to any input image regardless of its modality or contrast — CT, T1, T2, FLAIR, or otherwise. With probability 0.10 this step is skipped and a single global remap is applied instead (step 3 below, with $C = 1$).

2. Spatial sub-parcellation. Each intensity class C_i is further subdivided into spatial sub-regions by a Voronoi partition seeded with points sampled within C_i (Fig. 1, column 3 shows the combined effect of this step and the remap of step 3), giving sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are contiguous in space as well as similar in intensity. This second partition breaks the spatial correlation of a purely intensity-based grouping: two voxels with the same intensity in different parts of the anatomy can land in different sub-regions and receive independent contrast perturbations. Each class is subdivided into m_i sub-regions sampled from

$\{2, \ldots, 10\}$ with probability 0.60, and left as one region (no sub-split) with probability 0.40.

3. Signed affine remap. For each sub-region R, let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0,1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is a per-region affine function of the *real* intensity x_v, it preserves the relative structure of the signal within each region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the region-to-region contrast relationship is randomized, and $\alpha_R < 0$ additionally allows local contrast inversions that no acquisition protocol produces on its own, widening the space of contrasts the network sees beyond what any single real sequence could supply. Background voxels are zeroed at this step (and remain zero through the rest of the pipeline). The result is optionally Gaussian-blurred (a σ sampled from $\{0, 0, 0, 0.3, 0.5, 0.8\}$, weighted toward no blur).

4. Anatomical-label affine remap (optional). Steps 1–3 are unsupervised: the k-means/Voronoi partition groups voxels by intensity and spatial proximity, which need not coincide with the true anatomical boundary between two structures. When two adjacent tissues happen to share similar intensity at a given training iteration, the partition can place both sides of a real label border inside the same region, so step 3 assigns them the same target contrast and the boundary is, by chance, erased rather than randomized. This last step corrects for that: for each real anatomical label ℓ present in the volume, with probability 0.5 (skipped below 4 foreground voxels) we resample a fresh (μ_ℓ, α_ℓ) and apply Eq. (1) *again*, restricted to ℓ's voxels only, **on top of the step-3 output** – i.e. this affine remap acts on the already-remapped (step 1–3) intensities, not on the original voxel values, so it composes with rather than replaces the unsupervised remap (Fig. 1, column 4 shows the resulting per-label flat-remap value, computed prior to this step). Because the label assignment is independent of whatever k-means/Voronoi region a voxel fell into, this step reintroduces a full-strength contrast discontinuity exactly at the true label border regardless of whether the unsupervised partition happened to preserve it – sharpening the boundary cue the segmentation network can rely on, rather than sharpening the underlying anatomy. The step only ever consumes the segmentation labels the network is already being trained to predict – whatever labels happen to be annotated in the training set, including sparse or partial annotations – so it requires no additional segmentation effort beyond what training already assumes. It is also optional: it can be disabled entirely (`label_remap_prob`$= 0$), in which case PALETTE degrades gracefully to steps 1–3 alone, e.g. on an unlabelled dataset. A final foreground z-score normalisation is applied after an optional second blur pass (same σ distribution as step 3), giving the training input shown in the last column of Fig. 1.

Per-step stochasticity, summarised: step 1 is replaced by a single global remap with probability 0.10; each intensity class is Voronoi-split with probability 0.60 (else left as one region); step 3 is always applied to whatever partition results from steps 1–2; step 4 is applied independently per label with probability 0.5.

3.2. Training protocol

PALETTE is applied on-the-fly as one contrast-augmentation operator inside an existing GPU-accelerated nnU-Net augmentation pipeline [12, 20], alongside that pipeline's standard spatial and artifact augmentations; it adds no learned parameters and no measurable training-time overhead. A single configuration – the same k-range, sub-region count, blur schedule, and α range given in §3.1 – is tuned once and reused unchanged across every dataset and modality in Sec. 4; no dataset receives its own hyperparameter sweep.

3.3. Causal ablation: isolating texture preservation

PALETTE differs from label-generative synthesis (e.g. SynthSeg) along two axes at once: the partition it perturbs (unsupervised k-means + Voronoi vs. a supplied anatomical label map as the sole generative substrate) and the perturbation itself (an affine remap of real intensities vs. resampling from a per-label Gaussian mixture). Comparing PALETTE to SynthSeg directly therefore cannot attribute an advantage to either axis in isolation.

We construct a controlled variant, *noise-fill*, that holds the k-means/Voronoi partition and the per-region statistics (μ_R, α_R) from Eq. (1) exactly fixed, but fills each sub-region with samples drawn from a Gaussian with that region's $(\mu_R, \alpha_R\sigma_R)$ instead of remapping the real intensities inside it – i.e. the same partitioning scheme, with SynthSeg-style noise substituted for the affine remap. Because the partition and target statistics are identical to PALETTE's own, any difference in downstream performance between PALETTE and noise-fill isolates the one variable that changed: whether real texture is preserved or discarded within each region. We report this comparison separately for texture-defined structures (identified by fine internal gradients or low boundary contrast) and boundary-defined structures (identified by a sharp, high-contrast border) in Sec. 4, as the texture-preservation hypothesis predicts a widening gap specifically on the former.

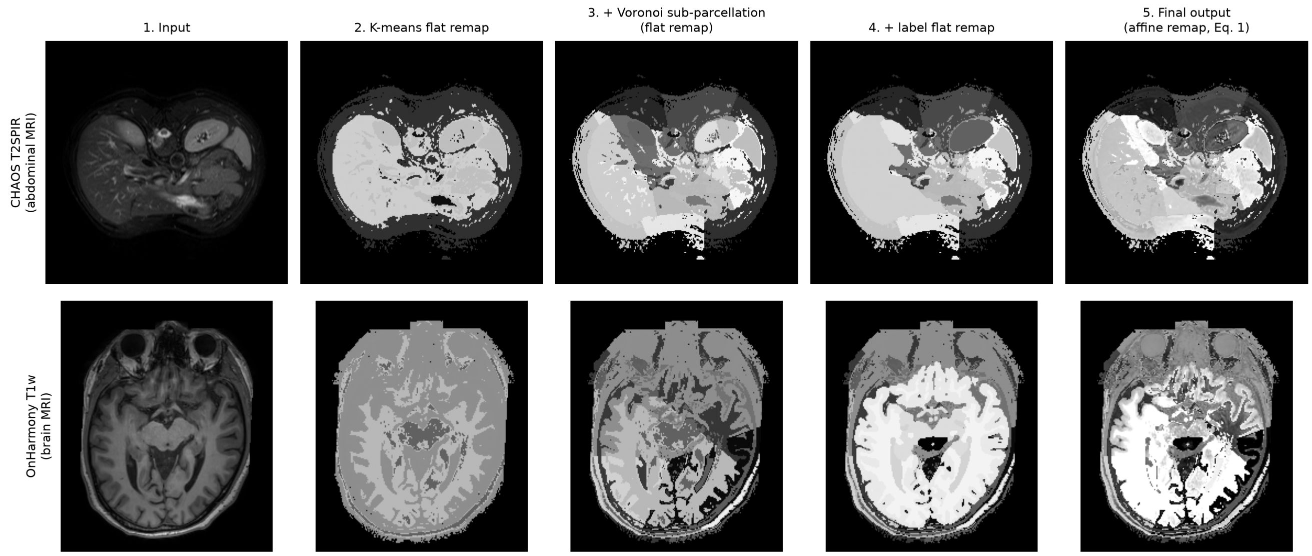

Figure 1. The PALETTE pipeline applied to one real CHAOS T2SPIR volume (top) and one real OnHarmony T1w volume (bottom). For display only, every panel is min–max normalised to $[0, 1]$ (the foreground z-score used at training time is never applied for visualisation); no panel is generated by an external model – every value shown is a deterministic function of the same real volume. Column 1: input. Column 2: *k-means flat remap* – each k-means class is painted with the precomputed target mean μ of Eq. (1) for its dominant (largest) Voronoi sub-region, i.e. the flat value that sub-region would take under the transform, not an average of observed pixel intensities. Column 3: the same flat-remap logic applied after Voronoi sub-parcellation, so each sub-region is painted with its own precomputed μ. Column 4: each real anatomical label is additionally painted with its own precomputed μ, overwriting the column-3 value on that label's footprint. Column 5: final PALETTE training input, obtained by applying the signed affine remap of Eq. (1) with texture (not a flat fill) at both the region level and, where labels are available, the anatomical-label level. Columns 2–4 isolate the flat target value chosen *before* the transform is applied with texture; column 5 shows the texture the transform actually produces once that value is used as its mean.

4. Experiments

4.1. Setup

Protocol. We train a single real-intensity source modality per dataset and evaluate on held-out modalities of the *same* segmentation task, using one fixed nnU-Net [12] configuration across all settings. No target-domain data, of any modality, is seen during training. Our full method, denoted **Ours** throughout, is PALETTE applied as one additional contrast-augmentation operator inside the Auglab pipeline [20]; the *Auglab* baseline is the same pipeline with the PALETTE operator removed, so the two differ by exactly the PALETTE transform. Every method is trained with the identical nnU-Net schedule — SGD with Nesterov momentum (0.99), the nnU-Net `polyLR` decay from an initial rate of 10^{-2}, 1000 epochs of 250 mini-batches, and the combined Dice-plus-cross-entropy loss — and differs only in the augmentation applied at the input; full configuration details are given in the supplementary.

Datasets. We evaluate on eight source-modality settings drawn from six datasets, spanning abdominal CT/MRI, brain MRI, and brain tumor MRI. In every setting the model is trained on the single source modality named below and tested on the held-out modalities named alongside it; no target modality is seen during training.

- **CHAOS** [14] (abdominal organ segmentation): train on either T1-in-phase or T2-SPIR MRI, and test cross-modality on the modalities *not* trained on — the complementary CHAOS contrasts (T1-in-phase, T1-out-of-phase, T2-SPIR) together with held-out CT from AMOS [13] and SLIVER07 [10].
- **ON-Harmony** [30] (brain segmentation, cross-contrast): train on either T1w or T2w, and test on the five held-out contrasts not trained on (the complementary structural contrast plus BOLD, DWI, EPI, and GRE). Reference labels were obtained on T1w with FreeSurfer [8] and registered to the other contrasts.
- **Open-MS** [16] (brain MS-lesion segmentation, cross-contrast): train on either FLAIR or T1w, and test on the held-out contrasts of the same lesion-segmentation task not trained on (FLAIR, T1w, T2w).
- **Brats2024-GLI** [7] (post-treatment glioma segmentation, cross-contrast): train on either T1n or T2w, and test on the held-out contrasts not trained on (T1c, T1n, T2-FLAIR, T2w).

Setting (train mod.)	Base	noEM	EM	Auglab	**Ours**
CHAOS T1in	21.9	65.3	85.2	85.4	**86.6**†
CHAOS T2spir	36.7	57.8	84.2	84.4	**86.0**†
ON-Harmony T1w	27.6	65.8	67.1	67.6	**68.5**†
ON-Harmony T2w	35.8	66.3	67.1	67.4	**67.8**†
Brats2024-GLI T1n	25.6	8.8	43.0	46.2	**46.6**
Brats2024-GLI T2w	18.1	8.7	43.8	46.6	**47.0**†
Open-MS FLAIR	25.2	4.6	32.8	32.8	**41.7**†
Open-MS T1w	16.7	4.0	35.1	35.8	**37.2**

Table 1. All-class Dice (%), averaged over folds and over all held-out test contrasts/datasets per setting (3-fold cross-validation). *Base* = no-augmentation; *noEM*/*EM* = the SynthSeg variants of Sec. 4.1; *Auglab* [20] is the strongest competing method; **Ours** = PALETTE applied on top of the Auglab pipeline. Best per row in bold. † marks settings where **Ours** beats Auglab by a case-level Wilcoxon signed-rank test with Holm correction ($p < 0.05$); this holds in six of eight settings (all except Brats2024-GLI T1n, $p = 0.64$, and Open-MS T1w, $p = 0.15$).

Baselines. We compare against four methods. (1) *No-augmentation*, the nnU-Net default training recipe. (2) *SynthSeg-noEM* [3], the original label-generative variant of SynthSeg (the implementation released on the authors' GitHub), which resamples intensities from a dense anatomical label map per training subject. (3) *SynthSeg-EM*, the variant described in the original work but never released publicly, which we re-implement; unlike SynthSeg-noEM it does not require a dense anatomical label map, but it is not label-free either — it is compatible with and relies on the sparse task labels naturally available in a segmentation dataset, which it uses to estimate its per-class intensity model. (4) *Auglab*, the published GPU-accelerated augmentation pipeline of Molinier et al. [20]. Auglab is a prior method in its own right, not an ablation of ours, and we report it among the competitors accordingly; our full method (Sec. 4.1) applies PALETTE *on top of* the Auglab pipeline, so the PALETTE-vs-Auglab comparison also functions as the add-one-component test isolating the contribution of the PALETTE transform.

Statistics. We report per-case paired Wilcoxon signed-rank tests [31] (not per-fold), with Holm correction [11] for multiple comparisons, so that significance reflects consistency across individual test cases rather than aggregate noise.

4.2. Main comparison

Every synthetic-contrast method vastly outperforms the no-augmentation baseline, confirming that some cross-contrast augmentation is indispensable under single-source evaluation: the baseline collapses whenever train and test contrast differ (e.g. 21.9 Dice averaged over the CHAOS T1in transfers, against 86.6 for Ours). SynthSeg-noEM, which resamples intensities from a dense anatomical label map, is competitive only on the brain-MRI settings where such maps are meaningful and collapses on the sparse-label tumor and lesion tasks (Brats2024-GLI, Open-MS); we return to this in Sec. 4.3. The re-implemented, label-using SynthSeg-EM is far stronger and closes most of that gap, but Ours beats it significantly in seven of eight settings.

The most informative comparison is against Auglab, the strongest competitor and the pipeline our full method builds on: the difference between the two columns is exactly the PALETTE transform. Ours is best in every setting and significantly better than Auglab in six of eight by a case-level Wilcoxon test. On most settings the aggregate margin is modest ($+0.4$ to $+1.6$ Dice) because both methods already saturate whole-organ boundaries, but on Open-MS FLAIR — a texture-defined lesion task — adding PALETTE lifts Dice from 32.8 to 41.7, a clear $+8.9$-point gain. This concentration of the advantage on texture-defined structures, rather than a uniform offset, is the effect we examine next.

4.3. Where the advantage concentrates

The advantage of PALETTE over Auglab is not uniform across structures. Consistent with the texture-preservation hypothesis in Sec. 3, the gap is widest on fine, texture-defined structures and on the most contrast-shifted test modalities, and narrow on whole-organ boundaries that survive any reasonable augmentation. The Open-MS FLAIR result above is the clearest case: MS lesions in FLAIR are defined by their internal signal/texture rather than by a sharp anatomical boundary, and it is exactly here that preserving real relative intensities inside each region — rather than overwriting them — yields the largest margin ($+8.9$ Dice). We report per-structure breakdowns for the settings where this effect is largest, so that it is verifiable case-by-case rather than asserted from the average.

4.4. Mechanism ablation: which components matter

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of the no-augmentation anchor (*Base*, Sec. 4.2): first the unsupervised k-means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 4); then the Voronoi spatial sub-parcellation (step 2); and finally the full signed affine remap of Eq. (1) (step 3), which together constitute PALETTE in its entirety and is denoted **Ours** in this ladder — identical to the standalone (no-Auglab) configuration used throughout Sec. 4.2. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE's own internal components from the separate question of whether it com-

Component	CHAOS	Open-MS	BraTS	ON-Harmony
Base (no augmentation)	36.1	25.2	23.8	34.8
+ k-means partition	84.8	30.6	–	–
+ label remap	87.7	34.6	–	–
+ Voronoi sub-parcellation	87.2	36.1	–	–
+ affine remap (**Ours**)	87.9	40.4	–	–
+ Auglab (deployed)	87.3	39.5	44.4	67.7

Table 2. Mechanism ablation (top five rows, additive, no Auglab) and deployment composability check (bottom row). Each rung adds exactly one PALETTE component on top of the row above; all-modality Dice (%), averaged over held-out test contrasts, folds 0–2, train fraction 50%. Intermediate k-means/label-remap/Voronoi rungs and the standalone **Ours** rung are available for Open-MS and CHAOS; BraTS and ON-Harmony currently report only Base and the deployed configuration, with the intermediate rungs and standalone **Ours** pending.

Method	FLAIR	T1w
PALETTE	**0.808**	**0.894**
Auglab	0.772	0.777
SynthSeg-EM	0.461	0.422
noise-fill	0.453	0.401
SynthSeg-noEM	0.332	0.319

Table 3. Texture fidelity measured by NGF similarity to the source scan (mean over n=30 Open-MS scans; higher = more source texture retained, $\approx 1/3$ = isotropic-noise floor). PALETTE preserves the most texture; the noise-based methods sit near the floor.

poses with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k-means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE, so stacking it on top of the ladder above is a composability/deployment check, not a further mechanism-ablation step. Tab. 2 keeps the two separate: the top five rows isolate PALETTE's internal mechanism, evaluated with no Auglab throughout, while the bottom row reports PALETTE plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

4.5. Causal ablation: noise-fill

PALETTE shares its k-means/Voronoi partitioning scheme with a much simpler procedure, so a natural question is whether the partitioning alone — not the texture-preserving intensity remap — is what helps. To isolate this, we compare against *noise-fill* (Sec. 3.3), which holds the partition and per-region statistics fixed and only replaces the real-intensity remap with label-style Gaussian noise; this makes noise-fill conceptually close to SynthSeg-EM, which likewise fills each region with sampled noise rather than preserving real texture. On Open-MS FLAIR — the texture-defined lesion task where the effect should be largest — PALETTE beats noise-fill by $\Delta = +2.24$ Dice (Holm-corrected $p = 0.0125$), winning on 17 of the 24 test cases against noise-fill's 7. Removing texture preservation while holding the partition and per-region statistics constant thus degrades performance specifically on this texture-defined setting, which is the outcome the texture-preservation hypothesis predicts. Structures whose boundaries are defined by sharp contrast, not fine internal texture, do not show the same degradation, isolating texture preservation — not the shared partitioning scheme — as the causal mechanism.

Direct measurement of texture fidelity. The Dice ablation above shows texture preservation *matters*; we also measure it directly. For each method we compute the normalized-gradient-field (NGF) similarity [9] between the augmented image and the source scan — an alignment-of-gradients measure that is high when local edge and texture structure is retained and near the isotropic-noise floor ($\approx 1/3$) when it is destroyed. Tab. 3 reports mean NGF over n=30 scans on the two Open-MS contrasts. PALETTE retains far more of the source texture than any noise-based method and significantly more than Auglab (FLAIR $p = 7.6 \times 10^{-3}$, T1w $p = 1.2 \times 10^{-6}$), while the label-style methods (SynthSeg-EM, noise-fill) and especially SynthSeg-noEM sit near the noise floor — exactly the ordering the mechanism predicts, and consistent with the Dice gaps above.

5. Discussion and Limitations

Aggregate margins are thin by design of the evaluation, not by accident. Every setting in Sec. 4 is single-source cross-contrast, the hardest regime in which to demonstrate a gain, and the strongest competitor is Auglab [20], a published pipeline that already includes a strong general-purpose augmentation stack. Adding the PALETTE transform on top of it improves aggregate Dice in all eight settings and does so significantly in six, across three anatomical regions and two modalities. We read the consistency of the direction of the effect, together with its concentration on texture-defined structures (Sec. 4.3), as more informative than any single aggregate margin.

The causal ablation isolates a mechanism, not a full explanation. The noise-fill comparison in Sec. 4.5 shows that texture preservation, specifically, drives the advantage on texture-defined structures; it does not by itself explain the smaller, more uniform gains observed on boundary-defined structures, which likely stem from the wider space of region-to-region contrast relationships (including the sign inversions in Eq. (1)) that no single real acquisition protocol supplies. Disentangling these two effects further is left to future work.

SynthSeg-noEM's collapse on sparse-label tasks is expected. The large gap between SynthSeg-noEM and the other methods on Brats2024-GLI and Open-MS (Tab. 1) is not an implementation artifact but a direct consequence of what SynthSeg-noEM requires. It generates images by resampling intensities from a *dense* anatomical label map covering the whole volume; on sparse-label segmentation tasks — a single glioma or a handful of MS lesions floating in an otherwise unlabelled brain — there is no such dense map, so the generated images carry almost no anatomical structure and the network cannot learn. SynthSeg-noEM is therefore expected to be competitive only where dense label maps are meaningful (the brain-MRI settings) and to collapse elsewhere; this is a property of the method's input requirements, not a caveat about our re-implementation.

Scope. PALETTE targets the single-source setting: one labelled training modality, no access to target-domain data of any kind. It is input-level and architecture-agnostic, and we do not claim it subsumes multi-source domain generalization methods that have access to several labelled source domains; the two are complementary rather than competing solutions to the same problem.

6. Conclusion

We introduced PALETTE, a label-free, single-source contrast augmentation that partitions the foreground of a scan by unsupervised k-means and Voronoi partition and applies a signed affine remap of the real intensities within each resulting sub-region, in contrast to label-generative synthesis methods that require a dense anatomical label map and replace real intensities with sampled noise. Across eight cross-contrast and cross-modality segmentation settings spanning abdominal, brain, and brain tumor imaging, a single nnU-Net configuration trained with PALETTE significantly outperforms a no-augmentation baseline and the original SynthSeg in every setting, and a re-implemented stronger SynthSeg-EM variant in seven of eight; added on top of the strongest competitor, the published Auglab pipeline, it improves all eight settings and does so significantly in six. A causal ablation that holds the partitioning scheme fixed and only swaps real-intensity remapping for label-style noise fill shows that texture preservation, not the shared partitioning scheme, drives the advantage on texture-defined structures – directly supporting the premise that motivated the method: that contrast-robust segmentation does not require discarding the real texture of the training image.

References

[1] Liviu Badea and Maria Popa. Toward generalizable multiple sclerosis lesion segmentation models. *arXiv preprint arXiv:2410.19623*, 2024. 3

[2] Jie Bao, Zhixin Zhou, Wen Jung Li, and Rui Luo. Structure-aware stylized image synthesis for robust medical image segmentation. *arXiv preprint arXiv:2412.04296*, 2024. 2

[3] Benjamin Billot, Douglas N. Greve, Oula Puonti, Axel Thielscher, Koen Van Leemput, Bruce Fischl, Adrian V. Dalca, and Juan Eugenio Iglesias. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. *Medical Image Analysis*, 86:102789, 2023. 1, 2, 6

[4] Vicent Caselles-Ballester, Eloy Martínez-Heras, Giuseppe Pontillo, Zoe Mendelsohn, Elena M. Marrón, Juan Luis García Fernández, Laia Subirats, Jon Stutters, Jeremy Chataway, Frederik Barkhof, Sara Llufriu, and Ferran Prados. TimeLesSeg: Unified contrast-agnostic cross-sectional and longitudinal MS lesion segmentation via a stochastic generative model. *arXiv preprint arXiv:2605.07955*, 2026. 3

[5] Stefano Cerri, Oula Puonti, Dominik S. Meier, Jens Wuerfel, Mark Muehlau, Hartwig R. Siebner, and Koen Van Leemput. A contrast-adaptive method for simultaneous whole-brain and lesion segmentation in multiple sclerosis. *NeuroImage*, 225:117471, 2021. 3

[6] Boqi Chen, Yuanzhi Zhu, Yunke Ao, Sebastiano Caprara, Reto Sutter, Gunnar Rätsch, Ender Konukoglu, and Anna Susmelj. Generalizable single-source cross-modality medical image segmentation via invariant causal mechanisms. *arXiv preprint arXiv:2411.05223*, 2024. 3

[7] Maria Correia de Verdier, Rachit Saluja, Louis Gagnon, Dominic LaBella, Ujjwal Baid, Nourel Hoda Tahon, Martha Foltyn-Dumitru, Jikai Zhang, Ahmed W. Moawad, et al. The 2024 brain tumor segmentation (BraTS) challenge: Glioma segmentation on post-treatment MRI. *arXiv preprint arXiv:2405.18368*, 2024. 5

[8] Bruce Fischl. FreeSurfer. *NeuroImage*, 62(2):774–781, 2012. 5

[9] Eldad Haber and Jan Modersitzki. Intensity gradient based registration and fusion of multi-modal images. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 726–733, 2006. 7

[10] Tobias Heimann, Bram Van Ginneken, Martin A Styner, Yulia Arzhaeva, Volker Aurich, Christian Bauer, Andreas Beck, Christoph Becker, Reinhard Beichel, György Bekes, et al. Comparison and evaluation of methods for liver segmentation from CT datasets. *IEEE Transactions on Medical Imaging*, 28(8):1251–1265, 2009. 5

[11] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. 6

[12] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. 2, 3, 4, 5

[13] Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhanng, Wanwan Ma, Xiang Wan, et al. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, pages 36722–36732, 2022. 5

[14] A. Emre Kavur, N. Sinem Gezer, Mustafa Barış, Sinem Aslan, Pierre-Henri Conze, Vladimir Groza, Duc Duy Pham, Soumick Chatterjee, Philipp Ernst, Savaş Özkan, et al. CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Analysis*, 69:101950, 2021. 5

[15] Pablo Laso, Stefano Cerri, Annabel Sorby-Adams, Jennifer Guo, Farrah Mateen, Philipp Goebl, Jiaming Wu, Peirong Liu, Hongwei Li, Sean I. Young, Benjamin Billot, Oula Puonti, et al. Quantifying white matter hyperintensity and brain volumes in heterogeneous clinical and low-field portable MRI. *arXiv preprint arXiv:2312.05119*, 2023. 3

[16] Žiga Lesjak, Alfiia Galimzianova, Aleš Koren, Matjaž Lukač, Koen Van Leemput, and Žiga Špiclin. A novel public MR image dataset of multiple sclerosis patients with lesion segmentations based on multi-rater consensus. *Neuroinformatics*, 16(1):51–63, 2018. 3, 5

[17] Da Li, Yongxin Yang, Yi-Zhe Song, and Timothy M. Hospedales. Learning to generalize: Meta-learning for domain generalization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018. 2

[18] Heng Li, Haojin Li, Jianyu Chen, Mingyang Ou, Hai Shu, and Heng Miao. RaffeSDG: Random frequency filtering enabled single-source domain generalization for medical image segmentation. *arXiv preprint arXiv:2405.01228*, 2024. 3

[19] Quande Liu, Cheng Chen, Jing Qin, Qi Dou, and Pheng-Ann Heng. FedDG: Federated domain generalization on medical image segmentation via episodic learning in continuous frequency space. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2

[20] Nathan Molinier, Hendrik Möller, Thomas Dagonneau, Anna Curto-Vilalta, Robert Graf, Matan Atad, Daniel Rueckert, Jan S. Kirschke, and Julien Cohen-Adad. One sequence to segment them all: Efficient data augmentation for CT and MRI cross-domain 3D spine segmentation. *arXiv preprint arXiv:2605.03098*, 2026. 2, 3, 4, 5, 6, 7

[21] Ju-Hyeon Nam, Nur Suriza Syazwany, Su Jung Kim, and Sang-Chul Lee. Modality-agnostic domain generalizable medical image segmentation by multi-frequency in multi-scale attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2

[22] Cheng Ouyang, Chen Chen, Surui Li, Zeju Li, Chen Qin, Wenjia Bai, and Daniel Rueckert. Causality-inspired single-source domain generalization for medical image segmentation. *IEEE Trans. Med. Imaging*, 42(4):1095–1106, 2023. 3

[23] Qiang Qiao, Wenyu Wang, Meixia Qu, Kun Su, Bin Jiang, and Qiang Guo. Medical image segmentation via single-source domain generalization with random amplitude spectrum synthesis. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3

[24] Sourya Sengupta, Satrajit Chakrabarty, Keerthi Sravan Ravi, Gopal Avinash, and Ravi Soni. SynthFM: Training modality-agnostic foundation models for medical image segmentation without real medical data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2

[25] Hyungseob Shin, Hyeongyu Kim, Sewon Kim, Yohan Jun, Taejoon Eo, and Dosik Hwang. SDC-UDA: Volumetric unsupervised domain adaptation framework for slice-direction continuous cross-modality medical image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

[26] Zixian Su, Kai Yao, Xi Yang, Qiufeng Wang, Jie Sun, and Kaizhu Huang. Rethinking data augmentation for single-source domain generalization in medical image segmentation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023. 3

[27] Franz Thaler, Martin Urschler, Mateusz Kozinski, Matthias A. F. Gsell, Gernot Plank, and Darko Stern. Semantic-aware random convolution and source matching for domain generalization in medical image segmentation. *IEEE Access*, 2025. 3

[28] Romain Valabregue, Ines Khemir, Eric Badinet, François Rousseau, Guillaume Auzias, and Reuben Dorent. SIAM: Head and brain MRI segmentation from few high-quality templates via synthetic training. *arXiv preprint arXiv:2605.02737*, 2026. 3

[29] Shujun Wang, Lequan Yu, Kang Li, Xin Yang, Chi-Wing Fu, and Pheng-Ann Heng. DoFE: Domain-oriented feature embedding for generalizable fundus image segmentation on unseen datasets. *IEEE Trans. Med. Imaging*, 39(12):4237–4248, 2020. 2

[30] Shaun Warrington, Andrea Torchi, Olivier Mougin, Jon Campbell, Asante Ntata, Martin Craig, Stephania Assimopoulos, Fidel Alfaro-Almagro, Karla L. Miller, Mark Jenkinson, Paul S. Morgan, and Stamatios N. Sotiropoulos. A multi-site, multi-modal travelling-heads resource for brain MRI harmonisation. *Scientific Data*, 12:595, 2025. 5

[31] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945. 6

[32] Zhenlin Xu, Deyi Liu, Junlin Yang, Colin Raffel, and Marc Niethammer. Robust and generalizable visual representation learning via random convolutions. In *International Conference on Learning Representations (ICLR)*, 2021. 3

[33] Ling Zhang, Xiaosong Wang, Dong Yang, Thomas Sanford, Stephanie Harmon, Baris Turkbey, Bradford J. Wood, Holger Roth, Andriy Myronenko, Daguang Xu, and Ziyue Xu.

Generalizing deep learning for medical image segmentation to unseen domains via deep stacked transformations. *IEEE Trans. Med. Imaging*, 39(7):2531–2540, 2020. 3

[34] Haoyu Zhao, Wenhui Dong, Rui Yu, Zhou Zhao, Du Bo, and Yongchao Xu. MoreStyle: Relax low-frequency constraint of fourier-based image reconstruction in generalizable medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2024. 3